

Matias Freire

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EDUCATION

Stevens Institute of Technology

Expected May 2028

B.S. in Computer Science

Hoboken, NJ

Coursework: 3D Computer Vision, Operating Systems, Algorithms, Data Structures, Real-Time and Embedded Systems, Computer Architecture, Systems Programming, Software Development Processes

TECHNICAL SKILLS

Programming: Java, Python, C/C++, JavaScript, SQL

Web Tools: REST APIs, TypeScript, React, Tailwind CSS, Next.js, Supabase, FastAPI

Libraries: PyTorch, Scikit-learn, Dash, Plotly, RTI Connex Python API, pandas, NumPy, LangGraph/LangChain

Developer Tools: Git, GitHub/GitLab, Docker, Jupyter Notebooks, Linux, RTI Admin Console

Certifications: Bloomberg Market Concepts, Hugging Face AI Agents Fundamentals

EXPERIENCE

AI/ML Fellow | AI Studio Fellow

May 2026 – Present

KPMG (via Cornell Break Through Tech)

Remote

- Developing a **multi-agent RAG research intelligence system** for **KPMG** that coordinates retrieval, reranking, synthesis, and verification agents to transform arXiv AI papers into cited, business-ready insights.
- Engineering the pipeline in Python, **LangGraph**, **PyPDF**, **Sentence Transformers**, **ChromaDB**, **Gemini**, and **RAGAS**, evaluating retrieval and generation using Recall@K, MRR, and faithfulness scoring.
- Completed an intensive AI/ML curriculum and built end-to-end projects using Python, scikit-learn, Keras, and LangChain, covering traditional ML, Deep Learning, RAG, Agentic AI and MCP

Software Engineering Intern

May 2026 – July 2026

GE Transportation, a Wabtec Company

Erie, PA

- Engineered a **real-time locomotive monitoring system** by containerizing a multi-service architecture with Docker in WSL and integrating RTI Connex DDS, FastAPI REST API endpoints, and Server-Sent Events to stream telemetry.
- Built the system's diagnostic capabilities using **condition-boundary analysis** and **scikit-learn-based classification**, while caching diagnostic indexes, history, and UI state in **Redis** for faster interaction.
- Made **900k+ samples reusable** across locomotive old and newer configurations by building Python/pandas signal and dataset pipelines and publishing them through an RTI Connex DDS Writer for playback testing.

Undergraduate Research Assistant | AI/ML Tools Intern

Aug 2025 – Aug 2026

Advanced Quantum Materials Lab

Hoboken, NJ

- AI Extraordinary Magnetoresistance Geometry Optimization:** Achieved a **test R2 of 0.90** and **MAE of 20.51** for sensitivity prediction by engineering COMSOL datasets, benchmarking XGBoost, Gaussian Process, MLP, Ridge, and Random Forest regressors, and building a Bayesian optimization pipeline to identify high-sensitivity sensor geometries.
- MoS2 Image Synthesis & Analysis Platform:** Unified three microscopy research workflows and able to generate **1,000+ synthetic samples in seconds** by engineering a Dockerized Dash and Plotly platform for configurable defect synthesis, preprocessing, and PyTorch analysis while also studying models like CNN, ResUNet, and MobileNetV2.

PROJECTS

FocusLab | Next.js, TypeScript, FastAPI, MCP, LangGraph, Docker, SQLite

Feb 2026 - Aug 2026

- Shipped an **in-app AI assistant** answering coursework questions and returning clickable lecture-slide downloads, grounding every answer in an **11-tool MCP server** over Canvas LMS and a **LangGraph ReAct agent** on Claude.
- Cut **note-to-LaTeX** writeup from manual retyping to one click, pipelining Claude vision into KaTeX so prose formulas and code screenshots come back as textbook-quality notation.
- Containerised the whole stack into **3 Docker Compose services** that start with one command, exposing **36 REST endpoints** over Canvas LMS and Spotify Web APIs with OAuth 2.0.

SHPE and Colorstack Stevens Chapter Website | Next.js, React, TypeScript, Supabase

Jan 2026 - May 2026

- Launched two responsive Next.js and React platforms **serving 250+ combined SHPE and ColorStack members** by owning **end-to-end development** with TypeScript, Tailwind, and Supabase.
- Accelerated full-stack development by using the **Supabase MCP** to implement **authentication, database-backed CRUD workflows**, and secure content management supporting member onboarding and sponsor outreach.

LEADERSHIP

Chief Web Officer | SHPE Stevens Chapter

March 2026 – Present

- Built and lead the chapter's Next.js/Supabase platform **supporting 150+ members**, while organizing technical events and sponsor-facing initiatives.

Vice-President | Former Head of External Relations | Colorstack Stevens Chapter

April 2025 – Present

- Drove **150% membership growth** and **120% higher** online engagement through **15+ events** per semester, partnerships, and technical programming.

Tutor-Counselor/Resident Assistant | STEP Bridge Summer Program

Jun 2025 – Aug 2025

- Mentored **52+ incoming freshmen** through Calculus/Physics instruction, personalized tutoring, and residential support.